

Few-Shot Object Detection for Apple Leaf Disease Severity Estimation Using Two-Stage Fine-Tuned YOLO

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Abstract—Disease severity estimation provides more actionable information than disease presence alone, yet robust estimation in orchard imagery remained challenging due to complex backgrounds, occlusion, and limited pixel-level annotations. This study investigated Few-Shot Object Detection (FSOD) for apple leaf disease severity estimation using a two-stage fine-tuned YOLO. The objective was to improve severity accuracy under complex field conditions while reducing dependence on large annotated datasets. Materials consisted of the Plant Pathology 2021 dataset, filtered to three classes (healthy, rust, and frog-eye leaf spot). A manually annotated subset (750 images) provided leaf and lesion masks to compute continuous severity as lesion area divided by leaf area multiplied by 100. Two pipelines were compared: (i) a single-stage approach performing leaf and lesion segmentation directly on full images, and (ii) a two-stage approach performing leaf detection, cropping the largest leaf region of interest, then applying leaf and lesion segmentation for severity computation. For FSOD, two-stage fine-tuning trained a base detector on healthy leaves and adapted to novel classes (rust and frog-eye leaf spot) under K-shot settings ($K = 5\text{--}50$), with sequential versus joint adaptation. Results showed that the two-stage severity pipeline reduced MAE from 1.49% to 0.66%. The best few-shot configuration achieved $mAP = 0.825$ and $mAP_{50-95} = 0.761$ at $K = 40$ and size medium under sequential fine-tuning. Further research could extend region selection beyond the largest leaf, evaluate instance-level shot definitions, and reduce annotation cost through weakly supervised or semi-supervised lesion segmentation.

Keywords—few-shot object detection; apple leaf disease; disease severity estimation; YOLO; two-stage fine-tuning.

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I. INTRODUCTION

Apple foliar diseases remain a persistent constraint in orchard management because infection reduces photosynthetic capacity, accelerates defoliation, and ultimately decreases yield and fruit quality [1]. Automated vision-based monitoring has therefore attracted sustained interest as an alternative to labor-intensive scouting. However, actionable crop protection decisions depend not only on disease identification but also on disease severity, which reflects infection progression and supports threshold-based interventions. Severity assessment in practice historically relied on manual visual estimation, often supported by diagrammatic scales and rater training. While these procedures improved consistency, visual scoring remained sensitive to rater experience, fatigue, and field conditions such as uneven illumination and partial occlusion, as documented in a century-spanning synthesis of visual severity estimation practices [2]. Broader analyses of plant

disease severity estimation further emphasized this transition from subjective manual rating toward algorithmic approaches and highlighted the need for robust and standardized evaluation [3], [4].

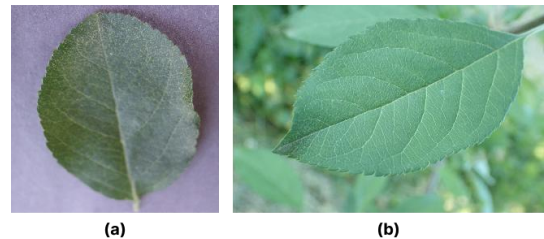


Fig. 1. Examples of publicly available datasets: (a) sample data with a simple background, and (b) sample data with a complex background.

Early computer vision pipelines typically began with disease classification using images acquired under controlled or semi-controlled conditions as shown in Fig. 1 (a), where

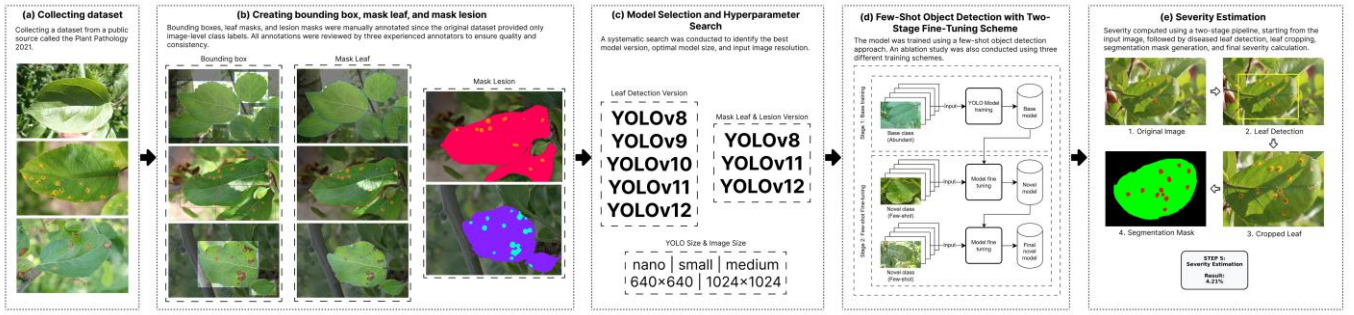


Fig. 2. Overview of the proposed research for estimating apple leaf disease severity in complex-background images. The pipeline combines few-shot object detection to localize leaves, detection-guided ROI cropping to focus on the target region, and subsequent leaf-lesion segmentation; the resulting segmentation outputs are then used to compute disease severity continuously based on the relative lesion area.

the leaf occupied most of the frame and background variability was limited [5], [6], [7], [8], [9]. Large-scale datasets with relatively homogeneous backgrounds enabled rapid adoption of deep convolutional models for accurate leaf-level recognition and facilitated mobile or edge deployment. For example, lightweight classifiers optimized for efficiency achieved high accuracy on controlled datasets and demonstrated feasibility for resource-constrained platforms [10]. Integrated frameworks built on controlled datasets similarly combined recognition with downstream severity-related reasoning or decision support [11]. These studies established a strong baseline, when background complexity was limited and lesions were visually salient, classification models provided stable disease recognition with high accuracy.

As deployment requirements shifted toward field scouting, research increasingly moved from controlled imagery toward complex background scenarios as shown in Fig. 1 (b). Under natural acquisition, leaves were captured with cluttered foliage, varying viewpoints, shadows, specular reflections, motion blur, and mixed disease symptoms. Models trained and tested on complex datasets demonstrated that performance depended strongly on generalization beyond controlled settings. A representative direction used complex orchard imagery for training and contrasted it with controlled imagery to evaluate robustness [1]. There are several studies that have made the transition from datasets with simple backgrounds to more complex backgrounds, such as in the study [12], [13], [14], [15] but there are still many shortcomings, such as only one leaf in the image or the leaves being picked leaves. More recent transformer-enhanced architectures also targeted natural-field variability explicitly, aiming to maintain performance when illumination and background changed substantially [16]. These studies supported the conclusion that classification accuracy in the laboratory did not directly translate to the field without architectural adaptations, data augmentation, or domain-specific training.

Classification alone remained insufficient when multiple leaves appeared in a single image, when symptoms were localized to limited regions, or when the goal included quantifying affected tissue. Consequently, research expanded from “what disease is present” toward “where symptoms occur,” motivating disease localization through object detection and segmentation. Lightweight YOLO-based pipelines were widely adopted because of their speed and suitability for near-real time inference. Optimized YOLO

variants demonstrated improved detection and recognition performance on standard plant disease datasets and small custom collections, often under relatively controlled imaging conditions where leaves and symptoms were visually separable [17]. In addition, end-to-end frameworks combined semantic leaf segmentation with disease classification, indicating that localization cues could support stronger diagnosis and interpretability [18].

The logical next step after localization was severity estimation, typically formulated as the proportion of infected tissue within the leaf. Under controlled or near-controlled conditions, severity estimation often relied on segmentation to delineate the leaf and lesion regions before computing an area ratio. Fine-tuned CNN models and hybrid pipelines reported strong accuracy for severity stage classification or severity quantification when images were captured in stable conditions, frequently with a single dominant leaf and limited occlusion [19]. Non-training or system-theoretic approaches also appeared, estimating severity through color discrepancy and recursive dynamic modeling, but these methods depended on reliable preprocessing assumptions (e.g., clean leaf segmentation) and careful parameterization [20]. Collectively, controlled-background severity studies demonstrated that accurate severity estimation was feasible when segmentation quality remained high.

Field deployment introduced new failure modes for severity estimation, background clutter reduced segmentation stability, overlapping leaves introduced ambiguous boundaries, and mixed symptoms complicated lesion delineation. Recent work therefore shifted toward architectures and pipelines designed specifically for natural field scenarios. Efficient segmentation networks aimed to balance speed and accuracy under real-world variability [21], while attention-optimized models improved lesion segmentation and achieved strong quantitative performance in complex settings [22]. Broader segmentation-based severity pipelines reported high agreement with reference methods across multiple crops and symptom types under non-uniform backgrounds [23].

For apple-specific severity estimation, a notable two-stage approach segmented leaves and lesions sequentially and reported high accuracy in complex backgrounds for frog-eye leaf spot, demonstrating the importance of separating leaf region understanding from lesion delineation [24]. Despite these advances, much of the field-focused severity literature remained centered on a single disease type or a constrained

severity labeling scheme, and many studies relied on extensive pixel-level annotations.

A persistent bottleneck across localization and severity estimation was the annotation burden, particularly for pixel-level lesion masks. Data scarcity became more severe when targeting rare diseases, specific orchards, or new symptom variants. Few-shot learning (FSL) addressed this constraint by adapting models from limited examples. In plant severity estimation, lightweight frameworks combined representation learning with few-shot strategies to estimate severity using minimal training samples, demonstrating the feasibility of reducing labeling requirements in apple leaf disease segmentation settings [25].

In parallel, few-shot object detection (FSOD) extended this concept to detection tasks, enabling adaptation to novel classes with limited annotations. Surveys and methodological advances clarified the FSOD landscape and emphasized challenges such as class imbalance, feature transferability, and catastrophic forgetting [26]. Recent models improved few-shot detection through enhanced classification representations, transformer-based prompting, and integration with foundation models [27], [28], [29]. Importantly, two-stage fine-tuning emerged as a practical strategy to decouple base learning from few-shot adaptation, supported by external memory mechanisms or staged optimization [30], and demonstrated effectiveness beyond natural images, including remote sensing settings [31]. Domain adaptation studies further suggested that foundation-model detection could be adapted to agricultural imagery under limited samples [32]. Overview of the proposed study can be seen in Fig. 2.

The literature indicated a clear progression as shown in Fig. 3, manual visual rating established operational relevance but suffered from subjectivity [2], classification under controlled backgrounds matured rapidly with efficient CNNs [10] and integrated systems [11], classification under complex backgrounds demanded stronger generalization strategies [1], [16], localization using YOLO-style detectors improved interpretability and multi-leaf handling, particularly under simpler acquisition conditions [17], [18], severity estimation under controlled backgrounds achieved high accuracy when segmentation remained stable [19], [20], severity estimation under complex backgrounds became feasible through advanced segmentation architectures and staged pipelines, including apple-specific work [24], yet remained annotation-intensive and frequently focused on a single disease type or constrained settings [21], [22], [23]. Meanwhile, few-shot approaches reduced labeling requirements, but the integration of FSOD-style adaptation with a practical severity pipeline in complex orchard imagery remained insufficiently explored [25], [26], [30], [31].

The present study addressed this gap by investigating apple leaf disease severity estimation in complex backgrounds across multiple disease classes using a YOLO-based pipeline designed for data efficiency. Severity was defined as a continuous percentage computed as lesion area divided by leaf area. Two severity pipelines were compared: (i) a single-stage approach that performed leaf and lesion segmentation directly on the original image prior to computing severity and (ii) a two-stage approach that first detected the dominant leaf, cropped the largest leaf region of interest, and then applied

leaf and lesion segmentation on the cropped region before computing severity. In addition, a two-stage fine-tuning protocol for few-shot detection was evaluated, training on a base class (healthy) and fine-tuning on novel disease classes (rust and frog-eye leaf spot) under a K-shot setup, with sequential versus joint fine-tuning ablations and a frozen-layer COCO-pretrained baseline. An example of the usefulness of calculating the estimated severity can be seen in Fig. 4. The contributions were summarized as follows:

- A continuous severity formulation based on lesion-to-leaf area ratio was adopted for apple leaf diseases under field imagery, enabling direct error-based evaluation.
- A systematic comparison between single-stage and two-stage severity pipelines quantified the effect of leaf detection and cropping in complex backgrounds.
- A two-stage fine-tuning FSOD protocol was evaluated for disease detection under limited samples, including sequential versus joint adaptation and comparison against a frozen-layer COCO-pretrained baseline.

Empirical evidence showed that the two-stage pipeline reduced severity estimation error relative to the single-stage approach and that two-stage fine-tuning improved detection performance under limited samples, supporting data-efficient deployment.

The remainder of this paper is organized as follows. Section II describes the materials and methods, including dataset selection, annotation protocol, model configurations, and evaluation metrics. Section III presents results and discussion, interpreting findings relative to prior work and practical constraints. Section IV concludes the paper and outlines implications and future research directions.

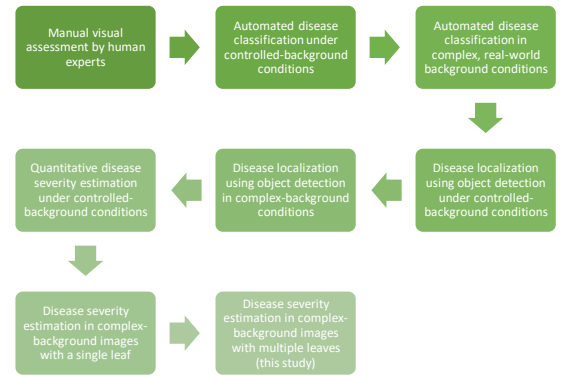


Fig. 3. Evolution of research approaches for plant disease analysis, illustrating the progression from manual visual assessment to complex backgrounds.

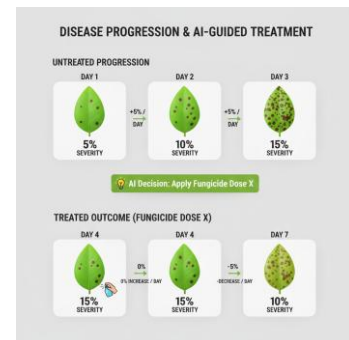


Fig. 4. An example illustrating the usefulness of estimating disease severity based on leaf image analysis. This visualization was generated by AI.

II. MATERIAL AND METHODS

A. Dataset and Data Selection

The study used the Plant Pathology 2021 dataset, which extended the Plant Pathology 2020 [33] collection and contained six leaf-condition classes: *healthy*, *scab*, *frog-eye leaf spot*, *complex*, *rust*, and *powdery mildew*. For the objectives of apple leaf disease severity estimation and few-shot adaptation, three classes were retained based on label availability and suitability for severity computation: *healthy*, *rust*, and *frog-eye leaf spot*. Fig. 5 shows examples of each class. Class counts in the filtered subset were as follows:

- Healthy: 4,624 images
- Frog-eye leaf spot: 4,352 images
- Rust: 2,077 images

The filtered dataset was randomly partitioned in a stratified manner by class into 70% training, 10% validation, and 20% testing. The resulting split sizes were:

- Training: 7,735 images
- Validation: 1,104 images
- Testing: 2,214 images



Fig. 5. Representative images from each disease class, showing healthy leaves (left), leaves affected by rust disease (center), and leaves exhibiting frog-eye leaf spot symptoms (right).

Because bounding box and pixel-level annotations for leaf and lesion masks were not provided by the source dataset, a manually annotated subset was created. A total of 250 images per class (750 images overall) were selected and uploaded to Roboflow for annotation. The annotated subset was partitioned into 70% training, 20% validation, and 10% testing, yielding:

- Training: 525 images
- Validation: 75 images
- Testing: 150 images

B. Annotation Protocol and Quality Control

All selected images were annotated in Roboflow. Three types of ground truth were prepared to support the proposed pipelines:

- 1) Bounding box annotations, used to detect all leaves in each image and assign the corresponding class label.
- 2) Leaf mask annotations, used to delineate the full leaf area for severity computation.
- 3) Lesion mask annotations, used to delineate diseased regions for severity computation.

Annotations were verified by three independent reviewers, each with experience in annotating bounding boxes and segmentation masks to reduce subjectivity and to improve consistency across disease types and background complexity. Review feedback was incorporated through iterative correction until masks and region boundaries were deemed consistent with the labeling guidelines. Fig. 6 presents examples of the bounding boxes created for each class, Fig. 7 shows examples of the leaf masks produced for each class,

and Fig. 8 shows examples of the lesion masks produced for each class.



Fig. 6. Examples of bounding-box annotations for each class, illustrating healthy leaves (left), leaves affected by rust disease (center), and leaves exhibiting frog-eye leaf spot symptoms (right).



Fig. 7. Examples of leaf segmentation masks for each class, showing healthy leaves (left), leaves affected by rust disease (center), and leaves exhibiting frog-eye leaf spot symptoms (right).



Fig. 8. Examples of lesion segmentation masks for diseased classes, illustrating rust lesions (left) and frog-eye leaf spot lesions (right).

In addition to qualitative examples, the overall characteristics of the annotated dataset were quantitatively analyzed. Fig. 9 illustrates the distribution of object annotation counts per image, showing that the majority of images contain only one to two annotated leaves, while images with a higher number of objects occur less frequently, reflecting variability in object density under complex background conditions. Furthermore, Fig. 10 summarizes the distribution of annotations across classes and dataset splits (training, validation, and test), highlighting class-wise differences in annotation availability as well as the consistency of data partitioning used for model development and evaluation.

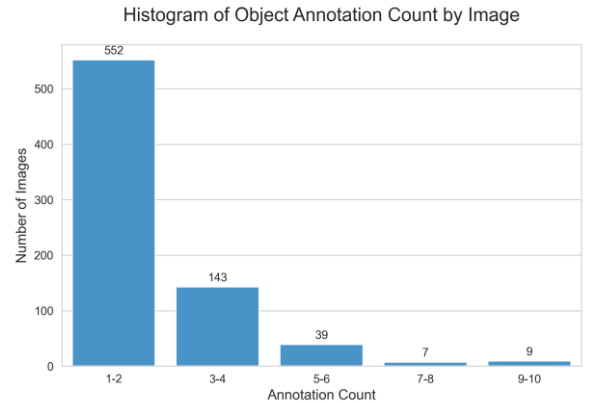


Fig. 9. Histogram of the number of object annotations per image in the dataset. The distribution shows that most images contain one to two annotated objects, while progressively fewer images include higher numbers of objects, indicating an imbalance in object density across the dataset.

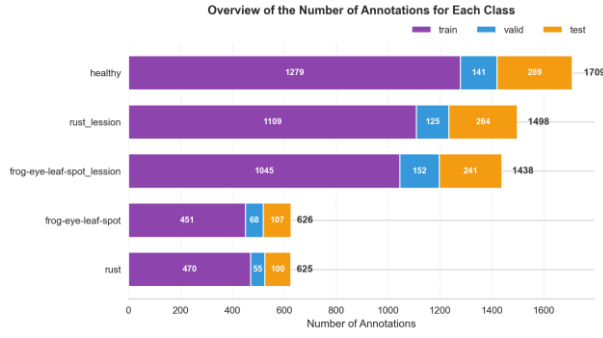


Fig. 10. Distribution of annotation counts for each class across the training, validation, and test splits. The stacked bars illustrate the relative contribution of each data split, highlighting differences in sample availability among healthy leaves, disease classes, and lesion annotations within the dataset.

C. Severity Definition

Ground-truth (GT) severity values were derived directly from the manual annotations. After the annotation process in Roboflow (bounding boxes, leaf masks, and lesion masks), the annotated masks were used to compute pixel areas for each image. Specifically, the leaf area and lesion area were calculated as the number of pixels belonging to the leaf mask and lesion mask, respectively, after all preprocessing steps (including any resizing) were applied consistently.

The computed areas and severity values were then exported and stored in a structured CSV file to ensure traceability, reproducibility, and ease of downstream analysis. Each record contained (i) an index, (ii) the image filename, (iii) the leaf area, (iv) the lesion area, and (v) the resulting severity percentage. An excerpt of the CSV structure is shown at TABLE I. below:

TABLE I.
EXAMPLE CSV STRUCTURE USED FOR GROUND-TRUTH (GT) SEVERITY
COMPUTATION (AREA COMPUTED IN PIXEL)

No	File_name	Leaf_area	Lesion_area	Severity
1	000001_Frog-eye-leaf-spot	1903944	39870	2.09%
2	000002_Frog-eye-leaf-spot	4243697	20146	0.47%
...	...	...	...	...
...	...	...	...	...
525	09167_Rust	1866073	18383	0.99%

Because GT severity was computed from manually produced segmentation masks, its accuracy depended primarily on annotation quality. The annotations underwent a quality review by three independent reviewers with experience in bounding-box and segmentation-mask annotation, followed by iterative refinement. Therefore, the resulting GT severity values were treated as reliable references for evaluating severity estimation performance. Disease severity was defined as a continuous percentage computed from segmentation masks:

$$\text{Severity} = \frac{\sum_{i=1}^K |L_i|}{|D|} \quad (1)$$

where L_i denoted the lesion area and D denoted the leaf area. Areas were computed as pixel counts within the corresponding masks, with any resizing operations applied

consistently across images prior to area calculation. Fig. 11 show example of severity computation.

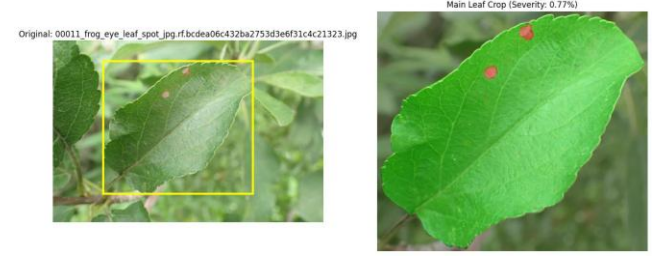


Fig. 11. An example illustrating the disease severity computation for the dominant leaf within an image, where severity is quantified based on the ratio between the segmented lesion area and the corresponding leaf area.

D. Severity Estimation Pipelines

Two severity estimation pipelines were implemented and compared. Fig. 12 and Fig. 13 illustrates the overall workflow of the proposed pipeline.

1) Single-stage pipeline (segmentation-only)

- Leaf segmentation and lesion segmentation were performed directly on the original image.
- Severity was computed from the resulting leaf and lesion masks using the area ratio.

2) Two-stage pipeline (detection-guided segmentation)

- Leaf detection was applied first to localize leaf candidates.
- The leaf region associated with the largest detected bounding-box area was selected as the primary region of interest.
- The selected region was cropped from the original image and used as input for leaf and lesion segmentation.
- Severity was computed from the masks predicted on the cropped region.

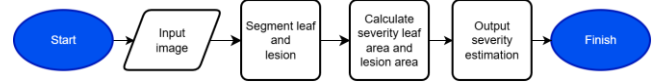


Fig. 12. Illustration of the single-stage pipeline (segmentation-only), in which disease severity is directly estimated from leaf and lesion segmentation results without an explicit leaf detection or region-of-interest cropping stage.

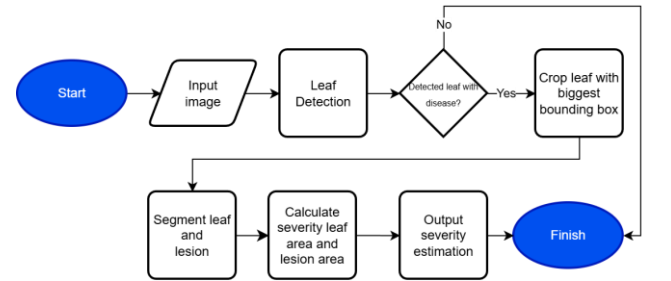


Fig. 13. Illustration of the two-stage pipeline (detection-guided segmentation), where leaf detection is first performed to localize regions of interest, followed by leaf and lesion segmentation for subsequent disease severity estimation.

E. Detection and Segmentation Models

Two YOLO-based model families were evaluated, and the final detection and segmentation configurations were selected based on the experimental results reported in Section III. For leaf detection, YOLO v8 through v12 were compared,

followed by a size-level comparison within the selected version (nano, small, and medium) to assess accuracy–efficiency trade-offs for leaf localization. The number of parameters for each evaluated detection version and size is summarized in TABLE II. For leaf and lesion segmentation, YOLO versions that provide native segmentation support and remain feasible to train under the computational constraints of this study were selected for comparison, namely YOLOv8, YOLOv11, and YOLOv12 within each selected version, multiple model sizes were also evaluated. YOLOv9 was not included because the available hardware could not support reliable training at the standardized batch size of 16 used in this study. Although training may be possible with a batch size smaller than 16, using a different batch size would introduce an additional confounding factor and reduce the fairness of comparisons across versions and model sizes. YOLOv10 was excluded because it does not provide segmentation capability for the required leaf–lesion mask prediction task. TABLE III. reports the parameter counts (in millions) for the evaluated YOLO segmentation versions and model sizes. In this table, “–” indicates that the corresponding model size is not available for that YOLO version, whereas “*” indicates that the YOLO version does not support the segmentation task and therefore cannot be trained for segmentation.

TABLE II.
MODEL PARAMETER COUNTS (IN MILLIONS) FOR THE EVALUATED YOLO
DETECTION VERSIONS AND SIZES

YOLO	v8	v9	v10	v11	v12
Nano/Tiny	3.2	2.0	2.7	2.6	2.6
Small	11.2	7.2	8.1	9.4	9.3
Medium	25.9	20.1	16.5	20.1	20.2

TABLE III.
MODEL PARAMETER COUNTS (IN MILLIONS) FOR THE EVALUATED YOLO
SEGMENTATION VERSIONS AND SIZES

YOLO	v8	v9	v10	v11	v12
Nano	3.4	-	*	2.9	2.8
Small	11.8	-	*	10.1	9.9
Medium	27.3	-	*	22.4	22.5
Compact	-	27.9	*	-	-
Extensive	-	60.5	*	-	-

F. Two-Stage Fine-Tuning for Few-Shot Detection

Few-shot adaptation followed a two-stage fine-tuning protocol:

- Stage 1 (base training): training was conducted using the base class, defined as *healthy*.
- Stage 2 (novel adaptation): the model was fine-tuned on novel disease classes: *rust* and *frog-eye leaf spot*.

A K-shot design was applied, where K denoted the number of labeled samples per class and ranged from 5 to 50 in increments of 5 (e.g., the 5-shot setting used 5 labeled samples for each included class). Three training schemes were evaluated for comparison. Scheme 1 performed sequential adaptation from the base class to novel classes (base → rust → frog-eye leaf spot), and its workflow is illustrated in Fig. 14. Scheme 2 adapted to the novel classes jointly (base → rust & frog-eye leaf spot), as shown in Fig. 15. Scheme 3 trained

the YOLO model directly on healthy, rust, and frog-eye leaf spot under the same K-shot sampling strategy to provide an additional comparison.

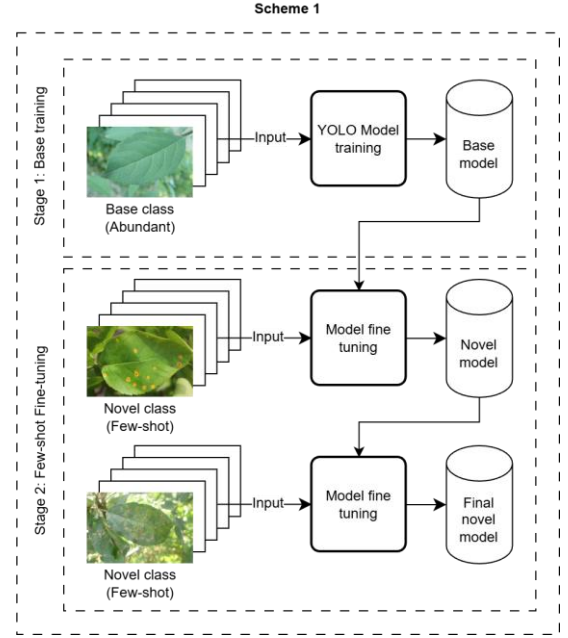


Fig. 14. Fine-tuning Scheme 1: sequential two-stage adaptation under the K-shot protocol, where the base model is first fine-tuned on the rust class and subsequently fine-tuned on the frog-eye leaf spot class (base → rust → frog-eye leaf spot).

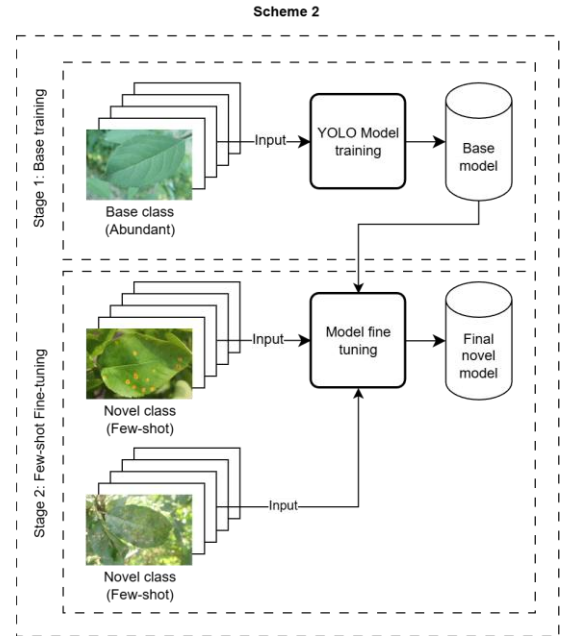


Fig. 15. Fine-tuning Scheme 2: joint two-stage adaptation under the K-shot protocol, where the base model is fine-tuned once using combined data from rust and frog-eye leaf spot classes (base → rust & frog-eye leaf spot).

G. Training Setup and Implementation Details

All models were trained using the same hardware and YOLO model configuration, which includes an AMD Ryzen 7 8700F processor, 32GB of RAM, and an RTX 5070 Ti GPU with 16GB of VRAM. The detailed YOLO configuration can be seen in TABLE IV. and the research flow from data preparation, model experiment, to few-shot experiment can be seen in Fig. 16

TABLE IV.
YOLO TRAINING CONFIGURATION USED IN THIS STUDY

No	Argument	Value
1	Epoch	100
2	Batch size	16
3	Image size	640 and 1024
4	Device	GPU
5	Worker	8
6	Pre-trained	True
7	Optimizer	ADAM

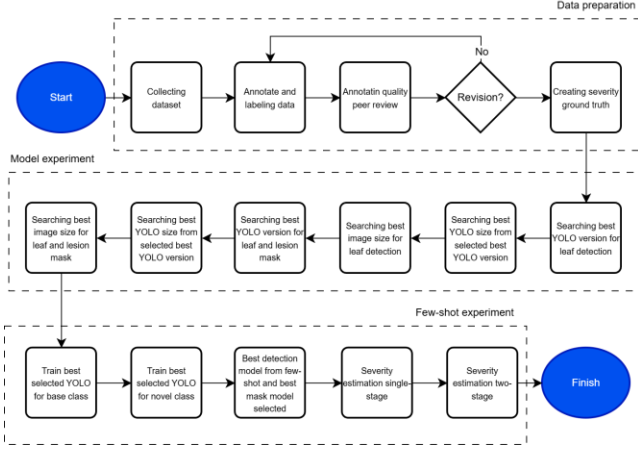


Fig. 16. Research workflow illustrating the process from data preparation and model experimentation through to the few-shot learning experiments.

H. Evaluation Metrics

Detection performance was first evaluated on the held-out test set using mAP50 and mAP50–95. The same metrics were then used to evaluate the segmentation models on the test set to ensure consistent comparison criteria across tasks. Next, few-shot detection performance was reported using mAP50 and mAP50–95 for each K-shot configuration ($K = 5-50$), enabling analysis of how detection accuracy changed as the number of labeled samples per class increased. Finally, severity estimation performance was assessed using Mean Absolute Error (MAE), computed between the predicted continuous severity and the ground-truth severity derived from manual leaf and lesion masks. MAE was selected instead of categorical severity levels as in research [24], since categorical grading could mask meaningful errors. For example, if Level 1 covered the range 0.1–1.0 and Level 2 covered 1.1–2.0, a ground-truth severity of 0.8 and a prediction of 0.2 would still fall within the same category and could be counted as correct, despite a large quantitative discrepancy. A continuous error metric such as MAE avoided this issue by directly penalizing the absolute difference between ground truth and prediction. The metric formulations used in this study are provided below:

$$\text{mAP50} = \frac{1}{C} \sum_{c=1}^C \text{rAP}_{\text{IoU}=0.50}^{(c)} \quad (2)$$

$$\text{mAP50-95} = \frac{1}{C} \sum_{c=1}^C \left(\frac{1}{10} \sum_{k=0}^9 \text{rAP}_{\text{IoU}=0.50+0.05k}^{(c)} \right) \quad (3)$$

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |S_i - \hat{S}_i| \quad (4)$$

III. RESULTS AND DISCUSSION

This section reported experimental results in a sequence: (1) selection of the most suitable YOLO version for leaf object detection, (2) selection of the most suitable YOLO version for segmentation (leaf and lesion masks) and the effect of input image size, (3) evaluation of few-shot object detection (FSOD) using two-stage fine-tuning with multiple training schemes and K-shot settings, and (4) the final comparison of single-stage versus two-stage severity estimation pipelines using the best-performing components.

A. YOLO Version and Size Selection for Leaf Object Detection

1) *YOLO version comparison (v8–v12)*: Leaf detection performance was first compared across YOLO versions v8 through v12. The evaluation used the test split and the primary selection criterion was the highest detection accuracy as measured by mAP (with mAP50 and mAP50–95 reported). The YOLO version comparison was conducted only using the small model size, as it provided the most suitable trade-off between inference speed and accuracy among the nano, small, and medium variants. As shown in Fig. 17, YOLO v12 achieved the strongest detection performance on the test set and therefore served as the detection backbone for subsequent experiments. The selection of YOLO v12 was consistent with the broader trend in plant-disease localization research that emphasized efficient detectors for field imagery while maintaining high precision under variability in illumination and background clutter [17].

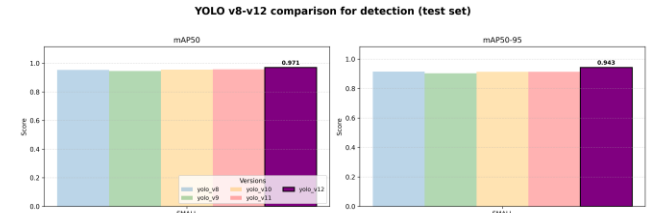


Fig. 17. Performance comparison of YOLOv8–YOLOv12 for leaf detection on the test set.

2) *YOLO v12 size comparison (nano, small, medium, large)*: After selecting YOLO v12, model size was evaluated to balance accuracy and practical deployment constraints. A size comparison was conducted among nano, small, medium, and large. As reported in Fig. 18, the nano model achieved a marginally higher mAP50 than the small model (0.972 vs 0.971), while the small model achieved a substantially higher mAP50–95 than the nano model (0.943 vs 0.932). In addition, the training curves in Fig. 19 indicated that the small variant consistently achieved the best mAP50 and mAP50–95 during training. Because the difference in mAP50 between nano and small was negligible ($\Delta = 0.001$) but the mAP50–95 gap was more pronounced, YOLO v12-small was selected as the detection model. Prior work suggested that improvements in higher-IoU performance (captured by mAP50–95) were particularly important when detection output guided subsequent processing steps, such as cropping and

downstream segmentation, because localization precision reduced ROI contamination by background pixels.

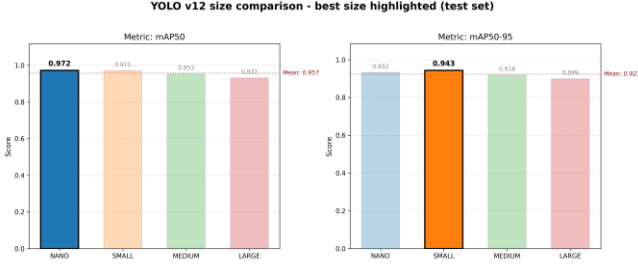


Fig. 18. Performance comparison of different YOLOv12 model sizes on the training set.

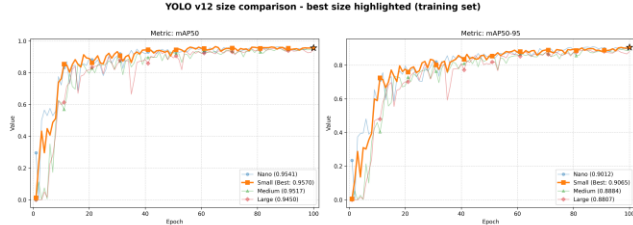


Fig. 19. Performance comparison of different YOLOv12 model sizes on the training set.

B. Segmentation Model Selection and the Effect of Input Image Size

1) *YOLO version comparison for segmentation:* A comparable selection procedure was applied to segmentation models. However, YOLO v9 was not included because its parameterization exceeded available training constraints and YOLO v10 did not support segmentation. Therefore, segmentation comparisons were conducted across YOLO v8, YOLO v11, and YOLO v12. As reported on Fig. 20, evaluation on the test split showed that YOLOv11 achieved the highest segmentation performance and thus served as the segmentation backbone for leaf and lesion mask prediction. The use of different versions for detection (YOLOv12) and segmentation (YOLOv11) reflected empirical performance differences and was maintained for subsequent experiments. Segmentation quality directly constrained severity accuracy in mask-based pipelines. Field-severity literature consistently reported that reliable lesion-area estimation depended on robust segmentation under complex backgrounds and variable symptom morphology [21], [23].

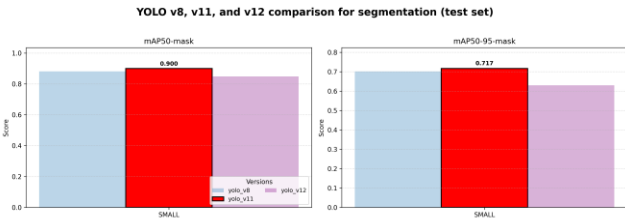


Fig. 20. Segmentation performance comparison of YOLOv8, YOLOv11, and YOLOv12 on the test set.

2) *YOLO version comparison (v8–v12):* A dedicated experiment examined the impact of input image size because lesion regions were visually small relative to the full leaf area (See Fig. 21 for example), a pattern commonly observed in rust and frog-eye leaf spot. Fig. 22 show performance changes

when input size increased from 640 to 1024. Three segmentation task settings were evaluated: Leaf-only segmentation, Lesion-only segmentation, Joint leaf-and-lesion segmentation. The results showed a mixed effect. For images without lesions (healthy), the 640 input size produced slightly higher performance than 1024. In contrast, lesion-focused settings benefited strongly from the larger input size. For lesion-only segmentation, mAP50–95 increased from 0.376 to 0.578 when image size increased from 640 to 1024. For joint leaf-and-lesion segmentation, mAP50–95 increased from 0.676 to 0.764. The improvement at higher resolution in lesion-related tasks was consistent with the hypothesis that small and fragmented lesion regions required higher spatial detail to avoid under-segmentation. Similar conclusions were reported in complex-background severity pipelines, where fine-grained lesion delineation determined the stability of area-based severity estimation [24]. Due to hardware constraints, the image-size experiment was conducted only with the nano segmentation model, the small and medium variants did not run as planned at 1024 resolution. This limitation was expected from a memory standpoint: increasing the input from 640×640 to 1024×1024 increased the number of pixels by $(1024^2/640^2) = 2.56 \times$, which increased activation and feature-map memory by approximately the same factor during training.

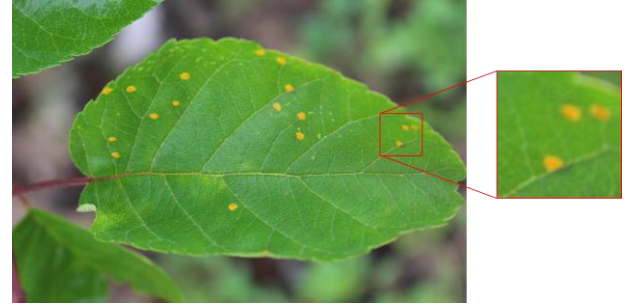


Fig. 21. Example image illustrating a case where the lesion area is relatively small compared to the total leaf area.

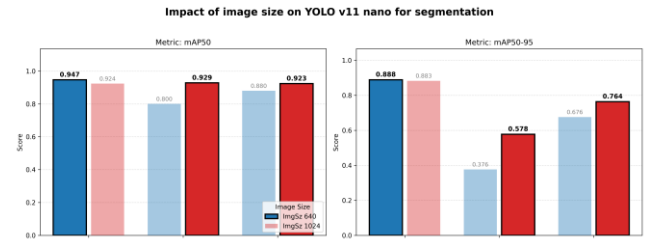


Fig. 22. Effect of input image size on the segmentation performance of YOLOv11-Nano on the test set across different experimental settings.

C. Few-Shot Object Detection Results Using Two-Stage Fine-Tuning

Few-shot detection experiments evaluated whether two-stage fine-tuning improved disease localization under limited labeled samples. The protocol trained YOLOv12 on the base class (healthy) and then adapted to the novel classes (rust and frog-eye leaf spot). K-shot settings ranged from $K = 5$ to 50 in increments of 5, where K denoted the number of labeled samples per class. As shown in Fig. 25, Scheme 1 produced the highest mAP50 and mAP50–95 among all schemes across model sizes. For the nano model, the peak performance

occurred at $K = 45$, achieving $\text{mAP50} = 0.868$ and $\text{mAP50-95} = 0.781$. The small model showed the same peak at $K = 45$, with $\text{mAP50} = 0.857$ and $\text{mAP50-95} = 0.768$. In contrast, the medium model reached its best performance with fewer samples at $K = 40$, achieving $\text{mAP50} = 0.848$ and $\text{mAP50-95} = 0.766$. The results indicated that two-stage fine-tuning was most effective when novel classes were introduced incrementally, consistent with FSOD findings that staged adaptation reduced interference between novel classes and supported more stable transfer from base representations [30], [31].

Fig. 25 reported mAP values observed during the training process, but additional identification was required to confirm generalization on the test set. After Scheme 1 was established as the best-performing scheme, a follow-up evaluation was conducted by re-testing Scheme 1 models on the held-out test set. As shown in TABLE V, the highest mAP50 on the test set was achieved by the medium model at $K = 40$ with $\text{mAP50} = 0.825$, whereas the highest mAP50-95 was achieved by the small model at $K = 35$ with $\text{mAP50-95} = 0.765$. This result indicated that although the best training performance was obtained at $K \geq 40$, the optimal K on the test set could be lower ($K = 35$), highlighting differences between training dynamics and test-set generalization.

TABLE V.
DETECTION PERFORMANCE OF YOLOv12 ACROSS DIFFERENT MODEL SIZES UNDER THE K-SHOT SETTING

Size	Best K-Shot	Precision	Recall	mAP50	mAP50-95
Nano	50	0.771	0.724	0.808	0.751
Small	35	0.774	0.777	0.817	0.765
Medium	40	0.781	0.728	0.825	0.761

D. Single-Stage vs Two-Stage Severity Estimation in Complex Backgrounds

The final evaluation compared severity estimation pipelines using the best-performing components identified above. Using the selected configuration (segmentation model: YOLO v11 Nano, FSOD training: Scheme 1, $K = 45$), the two-stage pipeline achieved a markedly lower severity error

than the single-stage pipeline. The MAE of the single-stage approach was 1.49, while the two-stage approach reduced MAE to 0.66 as shown in Fig. 24. The reduction in MAE supported the interpretation that detection-guided cropping reduced background contamination and improved segmentation stability under field conditions, Fig. 23 show the inference results from the best model using two-stage pipeline. This mechanism matched the logic of complex-background severity pipelines reported in the literature, where isolating the leaf region improved lesion delineation and downstream severity computation [24]. The result also extended prior severity estimation work by demonstrating improved performance in complex backgrounds and across multiple disease classes, rather than under controlled backgrounds or single-disease assumptions.



Fig. 23. Qualitative inference results obtained from the best-performing model configuration for detection and segmentation using the proposed two-stage pipeline.

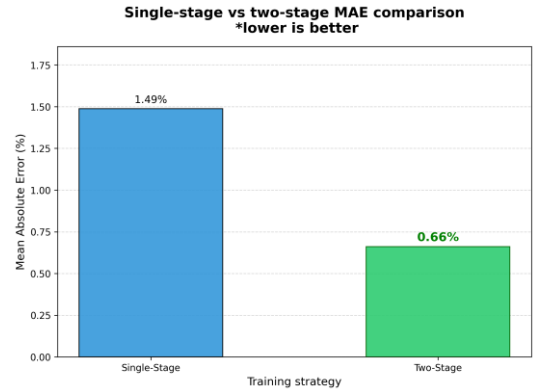


Fig. 24. Comparison of mean absolute error (MAE) between the single-stage and two-stage pipelines for disease severity estimation.

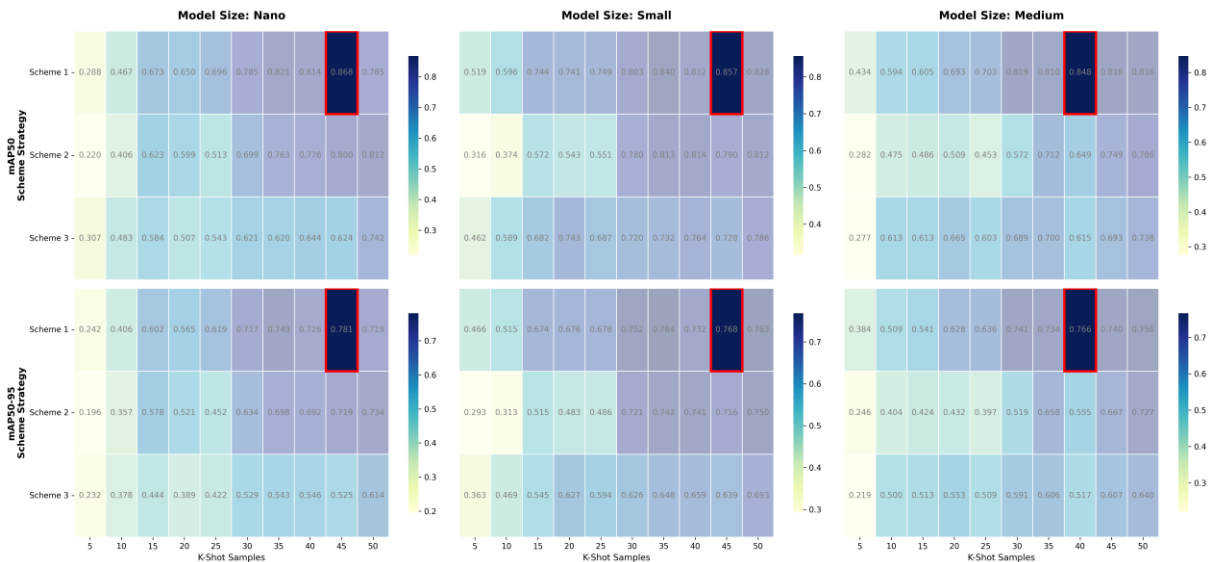


Fig. 25. Heatmap visualization of mAP50 and mAP50-95 scores for each model size (nano, small, and medium) across different K-shot settings.

E. Limitations and Implications

Several limitations remained. First, the two-stage severity pipeline selected the largest detected leaf as the ROI, which did not always correspond to the most symptomatic leaf when multiple leaves were present. Second, severity ground truth depended on manual masks, and although multi-review quality control improved consistency, pixel-level annotation remained time-intensive.

Despite these limitations, the results demonstrated that (i) careful YOLO model selection, (ii) resolution-aware segmentation for small-lesion conditions, and (iii) staged few-shot adaptation collectively improved the feasibility of apple leaf disease severity estimation in complex orchard imagery.

IV. CONCLUSION

Accurate severity estimation from orchard images remained difficult because complex backgrounds and limited annotations reduced the reliability of segmentation-based measurements. This study addressed the problem by combining YOLO-based detection, segmentation, and few-shot adaptation to estimate apple leaf disease severity as a continuous lesion-to-leaf area ratio. Detection-guided cropping improved robustness under field conditions, reducing severity error from MAE 1.49% (single-stage) to 0.66% (two-stage). Two-stage fine-tuning for few-shot disease detection further improved localization of novel classes (rust and frog-eye leaf spot), reaching mAP50 = 0.868 and mAP50–95 = 0.781 under sequential adaptation. These findings highlighted that ROI isolation and staged adaptation strengthened data-efficient severity monitoring in complex orchard imagery. Future research could extend ROI selection to multiple symptomatic leaves, refine few-shot protocols using instance-level sampling, and reduce annotation cost through weakly supervised or semi-supervised lesion segmentation, followed by validation across diverse orchards and acquisition devices. In addition, future work should evaluate newer object detection architectures such as the latest YOLO releases (e.g., YOLO26) as well as transformer-based detectors (e.g., DETR variants) to examine potential gains in localization robustness and generalization under complex orchard backgrounds.

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REFERENCES

- [1] M. A. Makruf, F. Sthevanie, and K. N. Ramadhani, "Classification of Apple Leaf Diseases Using a Modified EfficientNet Model," in *2024 International Conference on Intelligent Cybernetics Technology & Applications (ICICyTA)*, IEEE, Dec. 2024, pp. 960–965. doi: 10.1109/ICICyTA64807.2024.10913279.
- [2] C. H. Bock, K.-S. Chiang, and E. M. Del Ponte, "Plant disease severity estimated visually: a century of research, best practices, and opportunities for improving methods and practices to maximize accuracy," 2021, doi: 10.1007/s40858-021-00439-z/Published.
- [3] S. V. K and R. S. Kumar, "A Comprehensive Analysis on Severity Estimation of Plant Diseases by Deep Learning Models," in *2025 International Conference on Computational, Communication and Information Technology (ICCCIT)*, IEEE, Feb. 2025, pp. 240–245. doi: 10.1109/ICCCIT62592.2025.10927900.
- [4] T. Shi *et al.*, "Recent advances in plant disease severity assessment using convolutional neural networks," *Sci. Rep.*, vol. 13, no. 1, Dec. 2023, doi: 10.1038/s41598-023-29230-7.
- [5] S. Allaoua Chelloug, R. Alkanhel, M. S. A. Muthanna, A. Aziz, and A. Muthanna, "MULTINET: A Multi-Agent DRL and EfficientNet Assisted Framework for 3D Plant Leaf Disease Identification and Severity Quantification," *IEEE Access*, vol. 11, pp. 86770–86789, 2023, doi: 10.1109/ACCESS.2023.3303868.
- [6] B. Yang *et al.*, "A novel plant type, leaf disease and severity identification framework using CNN and transformer with multi-label method," *Sci. Rep.*, vol. 14, no. 1, Dec. 2024, doi: 10.1038/s41598-024-62452-x.
- [7] F. Qin, H. Wang, Q. Jiang, and H. Wang, "A novel method based on lesion expansion to assess plant disease severity," *Front. Plant Sci.*, vol. 16, 2025, doi: 10.3389/fpls.2025.1510663.
- [8] S. Verma, A. Chug, R. P. Singh, A. P. Singh, and D. Singh, "SE-CapsNet: Automated evaluation of plant disease severity based on feature extraction through Squeeze and Excitation (SE) networks and Capsule networks," 2022.
- [9] Y. Deng *et al.*, "An Effective Image-Based Tomato Leaf Disease Segmentation Method Using MC-UNet," *Plant Phenomics*, vol. 5, 2023, doi: 10.34133/plantphenomics.0049.
- [10] M. A. Makruf, F. A. D. Putra, A. R. Hidayah, and M. D. Sulistiyo, "Classification of Bean Leaf Lesions Using Modified EfficientNetV2 for Implementation in TensorFlow Lite," in *2025 International Conference on Data Science and Its Applications (ICoDSA)*, IEEE, Jul. 2025, pp. 381–386. doi: 10.1109/ICoDSA67155.2025.11157388.
- [11] L. D. S. D. and S. N. R. R., "An Integrated AI Framework for Tomato Leaf Disease Detection and Severity Assessment," in *2025 2nd International Conference on Electronic Circuits and Signaling Technologies (ICECST)*, IEEE, Oct. 2025, pp. 893–901. doi: 10.1109/ICECST66106.2025.11307455.
- [12] V. G. Bhujade and V. K. Sambhe, "Multi-Disease Classification and Severity Estimation of Cotton and Soybean Leaves Using Convolutional Neural Network," *Original Research Paper International Journal of Intelligent Systems and Applications in Engineering IJISAE*, vol. 2024, no. 11s, pp. 584–594, [Online]. Available: https://fieldcropnews.com/2016/07/soybean_disease_resource
- [13] Y. Li, D. Wang, C. Yuan, H. Li, and J. Hu, "Enhancing Agricultural Image Segmentation with an Agricultural Segment Anything Model Adapter," *Sensors*, vol. 23, no. 18, Sep. 2023, doi: 10.3390/s23187884.
- [14] T. Yang, S. Zhou, A. Xu, J. Ye, and J. Yin, "An Approach for Plant Leaf Image Segmentation Based on YOLOV8 and the Improved DEEPLABV3+," *Plants*, vol. 12, no. 19, Oct. 2023, doi: 10.3390/plants12193438.
- [15] K. R. P. Kumar, M. Gunasekar, K. Logeswaran, P. Dhanya Sree, P. Malathi, and V. Vignesh, "Fig Leaf Disease Prediction and Severity Estimation Using Deep Learning," in *2024 International Conference on Computing and Intelligent Reality Technologies (ICCIRT)*, IEEE, Dec. 2024, pp. 1–6. doi: 10.1109/ICCIRT59484.2024.10921856.
- [16] A. Iqbal, "PlantHealthNet: Transformer-Enhanced Hybrid Models for Disease Diagnosis and Severity Estimation in Agriculture," *IEEE Access*, vol. 13, pp. 101160–101176, 2025, doi: 10.1109/ACCESS.2025.3576990.
- [17] H. Wang, S. Shang, D. Wang, X. He, K. Feng, and H. Zhu, "Plant Disease Detection and Classification Method Based on the Optimized Lightweight YOLOv5 Model," *Agriculture*, vol. 12, no. 7, p. 931, Jun. 2022, doi: 10.3390/agriculture12070931.
- [18] K. Khan, R. U. Khan, W. Albattah, and A. M. Qamar, "End-to-End Semantic Leaf Segmentation Framework for Plants Disease Classification," *Complexity*, vol. 2022, 2022, doi: 10.1155/2022/1168700.
- [19] R. Kumar, A. Chug, and A. P. Singh, "Plant Leaf Diseases Severity Estimation using Fine-Tuned CNN Models," in *2023 6th International Conference on Information Systems and Computer Networks (ISCON)*, IEEE, Mar. 2023, pp. 1–6. doi: 10.1109/ISCON57294.2023.10111948.
- [20] D. Palma, F. Blanchini, and P. L. Montessoro, "A system-theoretic approach for image-based infectious plant disease severity estimation," *PLoS One*, vol. 17, no. 7 July, Jul. 2022, doi: 10.1371/journal.pone.0272002.
- [21] C. Zhao *et al.*, "Plant Disease Segmentation Networks for Fast Automatic Severity Estimation Under Natural Field Scenarios," *Agriculture (Switzerland)*, vol. 15, no. 6, Mar. 2025, doi: 10.3390/agriculture15060583.

- [22] K. Li, L. Zhang, B. Li, S. Li, and J. Ma, "Attention-optimized DeepLab V3 + for automatic estimation of cucumber disease severity," *Plant Methods*, vol. 18, no. 1, Dec. 2022, doi: 10.1186/s13007-022-00941-8.
- [23] J. P. Gonçalves, F. A. C. Pinto, D. M. Queiroz, F. M. M. Villar, J. G. A. Barbedo, and E. M. Del Ponte, "Deep learning architectures for semantic segmentation and automatic estimation of severity of foliar symptoms caused by diseases or pests," *Biosyst. Eng.*, vol. 210, pp. 129–142, Oct. 2021, doi: 10.1016/j.biosystemseng.2021.08.011.
- [24] B. Liu, H. Fan, Y. Zhang, J. Cai, and H. Cheng, "Deep learning architectures for diagnosing the severity of apple frog-eye leaf spot disease in complex backgrounds," *Front. Plant Sci.*, vol. 14, 2023, doi: 10.3389/fpls.2023.1289497.
- [25] P. Bedi, P. Gole, and S. Marwaha, "PDSE-Lite: lightweight framework for plant disease severity estimation based on Convolutional Autoencoder and Few-Shot Learning," *Front. Plant Sci.*, vol. 14, 2023, doi: 10.3389/fpls.2023.1319894.
- [26] Z. Xin, S. Chen, T. Wu, Y. Shao, W. Ding, and X. You, "Few-Shot Object Detection: Research Advances and Challenges," Apr. 2024, [Online]. Available: <http://arxiv.org/abs/2404.04799>
- [27] X. Jiang, Z. Li, M. Tian, J. Liu, S. Yi, and D. Miao, "Few-shot Object Detection via Improved Classification Features," in *2023 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, IEEE, Jan. 2023, pp. 5375–5384. doi: 10.1109/WACV56688.2023.00535.
- [28] A. Bulat, R. Guerrero, B. Martinez, and G. Tzimiropoulos, "FS-DETR: Few-Shot DETection TRansformer with prompting and without re-training," in *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, IEEE, Oct. 2023, pp. 11759–11768. doi: 10.1109/ICCV51070.2023.01083.
- [29] G. Han and S.-N. Lim, "Few-Shot Object Detection with Foundation Models," in *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, IEEE, Jun. 2024, pp. 28608–28618. doi: 10.1109/CVPR52733.2024.02703.
- [30] Y. Zhang, Z. Wang, S. Zhou, and Y. Zhao, "Faster Few-Shot Object-Detection: Two-Stage Fine Tuning Integrated with External Memories," in *2024 IEEE International Conference on Signal, Information and Data Processing (ICSIDP)*, IEEE, Nov. 2024, pp. 1–5. doi: 10.1109/ICSIDP62679.2024.10868816.
- [31] Z. Zhao, P. Tang, L. Zhao, and Z. Zhang, "Few-Shot Object Detection of Remote Sensing Images via Two-Stage Fine-Tuning," *IEEE Geoscience and Remote Sensing Letters*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2021.3116858.
- [32] R. Singh, R. B. Puhl, K. Dhakal, and S. Sornapudi, "Few-Shot Adaptation of Grounding DINO for Agricultural Domain," in *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, IEEE, Jun. 2025, pp. 5332–5342. doi: 10.1109/CVPRW67362.2025.00530.
- [33] R. Thapa, K. Zhang, N. Snively, S. Belongie, and A. Khan, "The Plant Pathology Challenge 2020 data set to classify foliar disease of apples," *Appl. Plant Sci.*, vol. 8, no. 9, Sep. 2020, doi: 10.1002/aps3.11390.